

Team Details:

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AI FAQ ASSISTANT API

Introduction

The AI FAQ Assistant is a robust, secure, and intelligent REST API platform designed to streamline FAQ creation, user authentication, and AI-powered content management. Built with Node.js and Express.js, the application utilizes MongoDB as its primary data store along with Mongoose ODM for elegant data modeling and schema validation.

Security is central to the architecture, implementing JWT (JSON Web Tokens) for stateless user sessions and route guarding alongside bcrypt for strong password encryption. The backend handles intensive operations like automated FAQ generation, semantic search queries, and dynamic context parsing. It also features robust centralized error handling and response status sanitization to prevent unhandled database leakage and ensure operational durability.

Project Overview

The **AI FAQ Assistant API** is an intelligent application designed to answer frequently asked questions automatically. Instead of manually searching through documents or asking a human support person, users can send a question to the API and receive a relevant answer.

The system provides a backend API that accepts user questions, processes them using AI/NLP techniques, searches the available FAQ information, and returns the most appropriate answer.

Main Objective

The main objective of this project is to develop an **AI-powered FAQ system** that can:

- Accept questions from users.
- Understand the user's question.
- Identify the relevant FAQ.
- Generate or retrieve an appropriate answer.
- Return the answer through an API.
- Reduce repetitive manual support work.

1. Brainstorming & Ideation Phase

1.1 Problem Identification

In many organizations, colleges, hospitals, companies, and online services, users repeatedly ask similar questions.

For example:

- What are the working hours?
- How can I register?
- How can I reset my password?
- What documents are required?
- How can I contact support?
- What is the admission procedure?

Answering the same questions repeatedly requires human effort and time.

Therefore, an automated FAQ assistant can be developed to provide answers quickly.

1.2 Proposed Solution

The proposed solution is an **AI FAQ Assistant API**.

The system allows an application or website to send a question to the API.

1.3 Idea

The main idea is to create a reusable API that can be connected to:

- Websites
- Mobile applications
- College portals
- Customer-support systems
- Chat applications
- Business applications

1.4 Innovation

The system improves traditional FAQ systems by allowing users to ask questions in natural language instead of searching manually through a list of questions.

Example

Traditional FAQ:

User searches: "Password Reset"

AI FAQ Assistant:

User asks: "I forgot my password. How can I change it?"

The system identifies the meaning and provides the relevant answer.

1.5 Expected Benefits

- Saves time.
- Reduces repetitive questions.
- Provides 24/7 assistance.
- Gives consistent answers.
- Easy to integrate with other applications.
- Reduces workload for support staff.
- Improves user experience.